

3 theorems from Local criteria for the unit equation and the asymptotic Fermat's last theorem

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The proofs depend on deep algebraic structures, Frey curves, and the Taniyama-Shimura theorem not yet formalized in Lean, hence the `sorry`.

Theorem 1 [1]. Let F be a totally real number field of degree n , and let $p \geq 5$ be a rational prime. Suppose:

1. $\gcd(n, p-1) = 1$;
2. 2 is either inert or totally ramified in F ;
3. p totally ramifies in F .

```
theorem theorem_1
  (h_real : IsTotallyReal F)
  (p : ℕ) [hp : Fact p.Prime] (hp_ge : p ≥ 5)
  (ha : Nat.gcd n (p - 1) = 1)
  (hb : Inert 2 F ∨ TotallyRamified 2 F)
  (hc : TotallyRamified p F) :
  AsymptoticFLT F := by
  sorry
```

Then the asymptotic Fermat's last theorem holds over F .

Theorem 2 [1]. Let F be a totally real number field of degree n . Suppose:

1. $n \equiv 1$ or $5 \pmod{6}$;
2. 2 is inert in F ;
3. 3 totally splits in F .

```
theorem theorem_2
  (h_real : IsTotallyReal F)
  (ha : n % 6 = 1 ∨ n % 6 = 5)
  (hb : Inert 2 F)
  (hc : TotallySplits 3 F) :
  AsymptoticFLT F := by
  sorry
```

Then the asymptotic Fermat's last theorem holds over F .

Theorem 3 [1]. Let F be a totally real number field of degree n . Suppose:

1. n is odd;
2. 2 totally ramifies in F ;
3. 3 totally splits in F .

```

theorem theorem_3
  (h_real : IsTotallyReal F)
  (ha : Odd n)
  (hb : TotallyRamified 2 F)
  (hc : TotallySplits 3 F) :
  AsymptoticFLT F := by
  sorry

```

Then the asymptotic Fermat's last theorem holds over F .

References

- [1] Nuno Freitas, Local criteria for the unit equation and the asymptotic Fermat's last theorem, 2020.